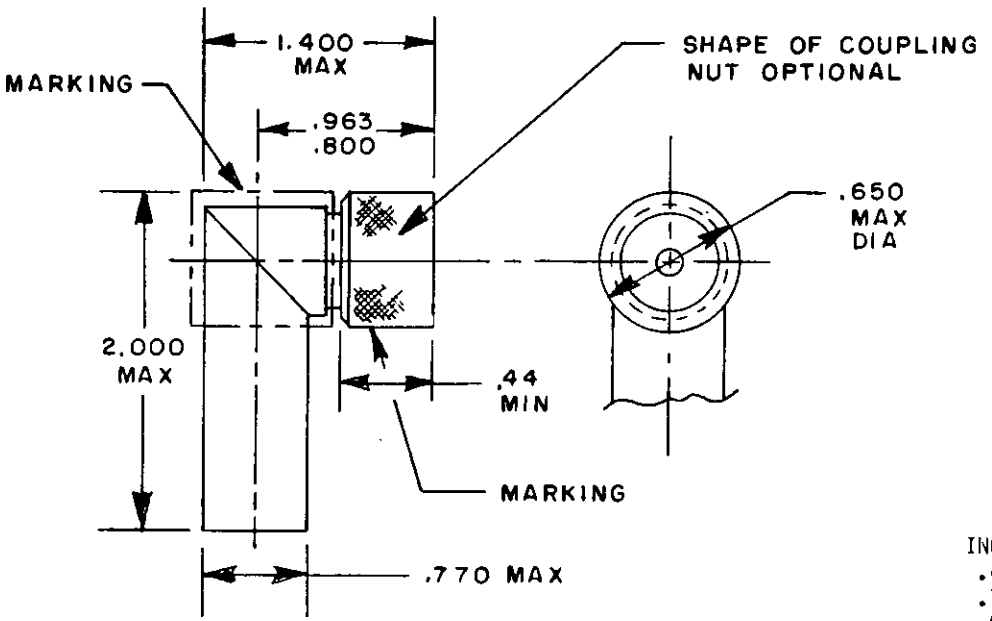
MILITARY SPECIFICATION SHEET

CONNECTORS, PLUG, ELECTRICAL, TRIAXIAL, RADIOFREQUENCY, (SERIES TRT (CABLED), PIN CONTACT, RIGHT ANGLE, CLASS 2)

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The complete requirements for acquiring the connectors described herein shall consist of this specification sheet and the latest issue of MIL-C-49142.



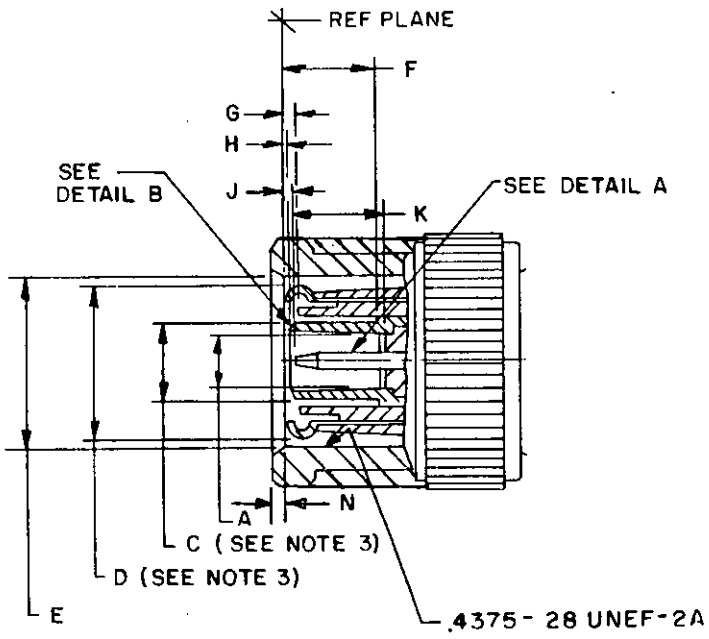
INCHES	MM
.44	11.18
.770	19.56
.650	16.51
.770	19.56
.800	20.32
.963	24.46
1.400	35.56
2.000	50.80

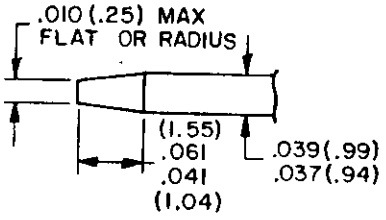
NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only.
3. Wrench flats are to accommodate standard wrench operings per FED-STD-H28.
4. 2.000 max defines the maximum length of the connector when assembled to the appropriate cable.
5. All undimensioned pictorial configurations are for reference purposes only.

FIGURE 1. General configuration.

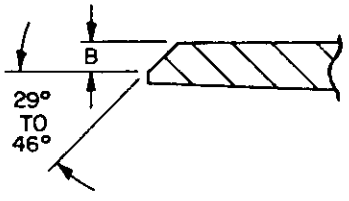
FSC 5935

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CENTER CONDUCTOR
DETAIL A



INTERMEDIATE CONDUCTOR
DETAIL B

Ltr	INCHES		MILLIMETERS	
	Min	Max	Min	Max
A	.123	---	3.12	---
B	.005	.010	.13	.25
C	.172	.178	4.37	4.52
D	.337	.347	8.56	8.81
E	.440	---	11.18	---
F	.213	.240	5.41	6.10
G	.001	---	.03	---
H	.007	.031	.18	.79
J	.018	.046	.46	1.17
K	.210	---	5.33	---
N	.063	---	1.60	---

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only.
3. Flared to meet mating characteristic test.
4. Metric equivalents are in parentheses or tabulated.

FIGURE 2. Mating dimensions for pin contacts.

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ENGINEERING DATA:

Nominal impedance: Non constant.

Frequency range: 0 to 500 MHz, minimum.

Voltage rating: 400 V rms, maximum working voltage at sea level. 100 V rms, maximum working voltage at 70,000 feet.

Temperature range: -65° to +165°C.

REQUIREMENTS:

Dimensions and configurations: See figures 1 and 2.

Force to engage and disengage:
Longitudinal force: Not applicable.
Torque: 2.5 inch-pounds, maximum.

Coupling proof torque: 15 inch-pounds minimum.

Inspection conditions: 15 inch-pounds, minimum.

Mating characteristics: See figure 2 for dimensions.
Intermediate contact:
Test ring ID - .169 inch maximum, 16 microinch finish.
Insertion force - 3 pounds maximum, when inserted a minimum of .093 inch.
Contacts with slotted members: Shall contact a .173 minimum diameter ring within .031 inch of their tip ends.
Outer contact:
Test ring ID - .319 inch maximum, 16 microinch finish.
Insertion force - 5 pounds maximum, when inserted a minimum of .093 inch.
Contacts with slotted members: Shall contact a .324 inch diameter ring within .031 inch of their tip ends.

Permeability: Applicable.

Hermetic seal: Not applicable.

Leakage: To be added.

Insulation resistance: 5000 megohms.

Inner conductor retention: 6 pounds minimum, axial force.

Dielectric withstanding voltage: At sea level, 1200 V rms, between center conductor and intermediate conductor. 500 V rms, between intermediate conductor and outer conductor.

Salt spray (corrosion): Applicable.

Vibration: Applicable.

Shock: Applicable.

Thermal shock: Applicable.

Moisture resistance: Applicable.

Conductor resistance: In milliohms, maximum.

	Initial	After environment
Center conductor	2.0	2.5
Intermediate conductor	0.5	0.6
Outer conductor	0.2	0.3

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TABLE I. Group Qualification.

Group	Submission and qualification of any of the following dash numbers	Qualifies the following dash numbers
I	-0001 -0002 -0005	-0001 -0002 -0005
II	-0003 -0004	-0003 -0004
III	-0006	-0006

Corona level:
Altitude: 70,000 feet.
Voltage: 250 V rms, minimum

RF high potential withstanding voltage:
800 V rms, between center conductor and intermediate conductor.
200 V rms, between intermediate conductor and outer conductor at 5 to 7.5 MHz.

Leakage current: Not applicable.

Cable retention force (for cable .200 to .325 outside dimension): 40 pounds, minimum.

Coupling mechanism retention force: 100 pounds, minimum.

Risetime degradation: To be added.

Connector durability: 500 cycles minimum, at 12 cycles per minute maximum.

Part number M49142/12-(dash number from table I).

TABLE I. Dash number, cross reference to cable.

Dash number	Cross reference number
0001	M17/134-00001
0002	M17/134-00002
0003	D3-7619-5/336
0004	D3-7619-5/338
0005	M17/116-RG307
0006 1/	M17/176-00002

1/ Not for Air Force use.

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Custodians:
Army - CR
Navy - EC
Air Force - 85

Review activities:
Army - AR, MI
Navy - OS, SH
Air Force - 11, 17, 99
DLA - ES

User activities:
Army - AT
Navy - AS, MC, SH
Air Force - 19

Agent:
DLA - ES

Preparing activity:
Army - CR

(Project 5935-3359-6)

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NOTE: This form may not be used to request copies of documents, nor to request waivers, deviations, or clarification of specification requirements on current contracts. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document*(s)* or to amend contractual requirements.

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE
MIL-C-49142/12	Connectors, Plug, Electrical, Triaxial, Radiofrequency, (Series TRT (Cabled), Pin Contact, Right Angle, Class 2)

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR
b. ADDRESS *(Street, City, State, ZIP Code)*	☐ USER
	☐ MANUFACTURER
	☐ OTHER *(Specify)*: _____

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

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